



P6

An Empirical Study of Observability and Identifiability of Hidden Kinematic Parameters: When Does an Estimator Beat the Mean Baseline

隐藏动力学参数的可观测性与可辨识性实证



An Empirical Study of Observability and Identifiability of Hidden Kinematic Parameters: When Does an Estimator Beat the Mean Baseline

Chinese Abstract

"从一段短运动轨迹反推隐藏物理参数"不是一个统一可解的问题：每个参数有自己的可观测性与可辨识性。本文在一个 96.8 万参数的小型世界模型（UDOS v7.0.3，单 CPU、固定 seed）上，用**可辨识性技能**（ $\text{identifiability skill} = 1 - \text{MAE_est} / \text{MAE_mean}$ ，相对均值预测器的归一化误差削减）逐参数给出能力边界。在固定历史窗 $W=6$ 、 $dt=0.5$ 下，四个隐藏参数的技能分化为：碰撞对速度 v_2 为 **0.820**（高度可辨识）、弹簧角频率 ω 为 **0.426**、初速度 v_0 为 0.318；而**加速度标量 a 为 -0.508**——**估计器 MAE 0.926 反而劣于均值基线 0.614，是一个诚实的负结果**。我们进一步定位根因：三维加速度向量本身可从速度对时间的最小二乘

斜率精确反演（技能=1.0），弱可辨识的只是“沿未知随机三维方向的带符号标量 a ”——方向翻转与符号反号给出同一轨迹，标量天然弱可辨识。该负结果在 hidden=64/128/256 三档模型尺寸上稳定（-0.523/-0.514/-0.508），不是模型容量伪影。本文主张：把控制论的可辨识性概念操作化为学习系统上的逐参数经验基准，并按参数诚实分级，而非笼统宣称“隐藏参数都能估计”。

Keywords (Chinese): identifiability; observability; system identification; world models; negative results; parameter estimation

Abstract

"Recovering hidden physical parameters from a short motion trajectory" is not uniformly solvable: each parameter has its own observability and identifiability. On a 0.97M-parameter small world model (UDOS v7.0.3, single CPU, fixed seeds), we score each hidden parameter by an **identifiability skill** ($1 - \text{MAE}_{\text{est}} / \text{MAE}_{\text{mean}}$, the normalized error reduction over a mean predictor). With a fixed window $W=6$ and $dt=0.5$, the four hidden parameters split sharply: the collision counterpart velocity v_2 is **0.820** (highly identifiable), the spring angular frequency ω is **0.426**, the initial velocity v_0 is 0.318; whereas the **signed acceleration scalar a is -0.508—the estimator MAE 0.926 is actually worse than the mean baseline 0.614, an honest negative result**. We localize the root cause: the 3D acceleration vector is exactly recoverable from the least-squares slope of velocity versus time (skill=1.0); what is weakly identifiable is only the signed scalar a along an unknown random 3D direction—flipping the direction and flipping (v_0, a) together yield the same trajectory, so the scalar is intrinsically weakly identifiable. The negative result is stable across model widths (hidden 64/128/256: -0.523/-0.514/-0.508), ruling out a capacity artifact. We argue for operationalizing control-theoretic identifiability as a per-parameter empirical benchmark on learned systems, graded honestly per parameter rather than claiming "all hidden parameters are estimable."

Keywords: identifiability; observability; system identification; world models; negative results; parameter estimation

Structured Abstract (background/problem → method → evidence/results → contribution)

- **Background and problem:** "recovering hidden physical parameters from a short motion trajectory" is not uniformly solvable—average error hides the fact that "some parameters are not identifiable at all from a short window, forced estimation only worse than the training-set mean"; in embodied/simulation systems, hidden parameter estimation errors amplify through time integration.
- **Method:** on a 0.97M small world model (UDOS v7.0.3, single CPU, fixed seeds, $\$W=6$, $\$dt=0.5$), scoring per parameter with the normalized metric **identifiability skill** = $1 - \text{MAE}_{\text{est}}/\text{MAE}_{\text{mean}}$ (error reduction over a mean predictor), and testing whether the negative result is a capacity artifact at three model widths (hidden 64/128/256); then root-cause localization with the "3D vector recoverable vs signed scalar" contrast.
- **Evidence and results:** the four parameters split—collision counterpart velocity v_2 skill **0.820**, spring ω **0.426**, initial velocity v_0 **0.318**; **counterintuitive negative result: signed acceleration scalar a skill = -0.508** (estimator MAE 0.926 worse than mean baseline 0.614). The negative result stable at three widths (-0.523/-0.514/-0.508), and the 3D acceleration vector inversion skill=1.0, root cause direction-sign inseparability, not the model failing to learn.

- **Contribution:** operationalizing control-theoretic identifiability as a per-parameter empirical benchmark on learned systems, honestly graded per parameter; taking the "negative result" as the protagonist selling point (not brushed past). **All numbers CPU fixed-seed single pilot point estimates**, no ≥ 30 seeds, no p-values/CIs; external industry numbers marked unverified.
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1 Introduction

Learned world models, when predicting, often implicitly "guess" physical quantities they have not explicitly observed: is this object moving uniformly, accelerating, pulled by a spring, or just collided? Once these hidden parameters are guessed wrong, long-horizon predictions systematically deviate. The common engineering practice is to train a network to estimate these parameters "end-to-end" from the window, then report an average error. But average error hides a key fact: **some parameters are not identifiable at all from a short window, forced estimation only worse than "directly using the training-set mean."**

Why is this an important problem? Because in embodied and simulation systems, hidden parameter estimation errors amplify through time integration: an acceleration estimate biased 0.5 is amplified by the quadratic $\frac{1}{2}at^2$ into significant position drift in a 4-step rollout. If this estimate itself is negative skill (worse than the mean), feeding it to the world model is only worse than not feeding. Hence "which parameters to learn, which to use analytic, which to not touch at all" is not academic fastidiousness but an engineering decision directly affecting downstream prediction error.

The question this paper answers is concrete: under fixed observation window $W=6$, sampling interval $dt=0.5$, how identifiable are each of the four hidden parameters—initial velocity magnitude v_0 , acceleration magnitude a , spring angular frequency ω , collision counterpart velocity v_2 ? Under what conditions does the estimator really beat "estimating nothing, directly using the mean"? We answer with a normalized metric **identifiability skill**: $\text{skill} = 1 - \frac{\text{MAE}_{\text{est}}}{\text{MAE}_{\text{mean}}}$, where MAE_{mean} is the "always predict the training-set mean" baseline. $\text{skill} > 0$ means the estimator beats the mean, $\text{skill} < 0$ means the estimator is worse (negative result), $\text{skill} = 1$ perfect estimation.

This metric's benefit is normalizing away "parameter scale": v_2 and a have different physical dimensions and value ranges, directly comparing MAE unfair; after dividing by the mean baseline, skill becomes a "relative gain" comparable across parameters. It also naturally encourages honesty: if the estimator learned nothing, skill approaches 0; if it learned wrong, skill goes negative—and the negative result is exactly this paper's differentiated contribution.

Choosing the mean baseline rather than stronger baselines (nearest neighbor, linear regression) is a deliberate conservative design. The mean baseline is the "do nothing" lower-bound reference: if a learned estimator cannot even beat the mean, it is engineering negative equity, to be turned off or replaced by analytic methods. Stronger baselines (e.g. least squares on acceleration) depress skill overall, but that is an "estimator vs analytic method" comparison answering another question. This paper first answers "do we want this learned estimator," the mean baseline most direct.

The evidence grade of external motivation must be stated. Recent embodied intelligence and world model literature heavily discusses "recovering physical parameters from vision/trajectories," "differentiable system identification," but related external numbers are mostly vendor or report caliber, **not independently verified in this study** (marked unverified, motivation only). We deliberately narrow scope to **synthetic kinematics with known ground truth**: uniform, uniform acceleration, simple

harmonic, 1D elastic collision four classes, state 6 dims. Known ground truth lets us compute "estimation error" per parameter, per sample, rather than only reporting one end-to-end prediction error.

The specific contributions of this paper:

1. **Per-parameter identifiability benchmark:** under a unified protocol giving skill, MAE, observability proxy for the four hidden parameters, including one **negative result** (acceleration scalar a skill=-0.508).
2. **Root cause localization:** distinguishing "the 3D acceleration vector exactly recoverable (skill=1.0)" and "the signed scalar a weakly identifiable along an unknown direction," attributing the negative result to the structural problem of **direction-sign inseparability**, not insufficient model capacity.
3. **Negative result robustness:** proving a 's negative skill stable at three model widths hidden=64/128/256 (-0.523/-0.514/-0.508), ruling out the "model too small, failed to learn" explanation.

Boundary statement: all numbers from a CPU fixed-seed single verification (`reports7/v7_verification.json`, `evidence_grade=verified`), **pilot point estimates**; no ≥ 30 -seed replication, no p-values/CIs, related positions retaining [RESULT NEEDED].

2 Related Work

Observability and identifiability are classic concepts of control theory and system identification [CITATION NEEDED: identifiability system identification Ljung; observability Kalman; structural identifiability Bellman Åström]. Observability asks "can state be inferred from observations," identifiability asks "can parameters be uniquely determined from data." Classic theory mostly targets linear/nonlinear models of known structure, giving analytic or algebraic criteria; learned world models push this problem to the new scenario "parameters in the network's implicit representation, structure unknown" [CITATION NEEDED: identifiability deep learning; identifiable neural networks; latent parameter estimation world models].

Three lines are most relevant. First, **differentiable system identification / physics-informed neural networks** [CITATION NEEDED: physics-informed neural networks; universal differential equations], writing physical residuals into the loss, often able to invert parameters; but under short windows, noise, unknown direction, whether parameters are uniquely identifiable still needs case-by-case analysis. Second, **latent variable estimation in world models** [CITATION NEEDED: latent dynamics estimation; world model latent parameter inference], mostly using VAE/state-space models to compress parameters into latents, then reporting reconstruction error. Third, **negative results / reproducibility movement** [CITATION NEEDED: negative results machine learning; reproducibility crisis], stressing honest reporting of failure cases.

Our positioning is **empirical benchmark + negative result**, no new identification algorithm claimed. Novelty is: operationalizing "identifiability" from a theoretical concept into a directly measurable skill metric on learned systems, honestly retaining a negative result. This complements P5 (conformal credibility): P5 cares "how wide the prediction interval," P6 cares "why the interval is wide—which parameters cannot be estimated at all."

Two often-confused concepts also need distinguishing. **Observability** cares "from an ideal observation of no noise, infinite length, can this quantity theoretically be computed"—in this paper, the acceleration vector's observability is 1 (state contains velocity, slope computable), while the signed scalar a 's observability is limited by direction-sign ambiguity. **Identifiability** cares "under the practical conditions

of finite window, finite sampling rate, noise, how much did estimation error reduce relative to baseline"—exactly what skill measures. A core observation here: **high observability does not equal high identifiability** (a's vector form observable, but scalar form skill negative), **high self-assessed confidence also does not equal high identifiability** (a self-assessed 0.798 but skill -0.508). Among these three is a gap we aim to fill: **the learned-systems literature rarely reports negative results per parameter, and even less often attributes negative results to structural non-identifiability rather than training failure**. Most papers report "our method successfully inverted on X% of parameters," but brush past failed parameters. This paper does the opposite: taking the failed parameter (a) as protagonist, proving at three model widths it is not training failure, then giving a structural explanation via vector/scalar contrast. This "taking failure as result" writing is more needed in engineering communities than theoretical—because what trips deployment is exactly these brushed-past parameters.

3 Problem Definition and Assumptions

3.1 Notation

State $s_t = [p_x, p_y, p_z, v_x, v_y, v_z] \in \mathbb{R}^6$. Observation window $X = (s_{t-W+1}, \dots, s_t)$, $W=6$, $\Delta t=0.5$. Hidden parameters $P \in \mathbb{R}^4$, slot order $(v_0, \text{accel_a}, \text{spring_omega}, \text{other_v2})$: v_0 initial velocity magnitude, accel_a acceleration magnitude, spring_omega spring angular frequency, other_v2 collision second particle velocity. Estimator $\hat{P} = f(X)$ output from the window by `SceneEstimator` (`scene.py`); mean baseline $\bar{P}$ the training-set that-slot mean.

The four parameters have different physical sources and identifiability mechanisms. v_0 is the initial velocity of uniform/accel trajectories, directly equal to observed velocity in uniform windows; accel_a the constant acceleration magnitude, theoretically given by velocity slope but with direction-sign ambiguity; spring_omega the simple harmonic angular frequency, determined by position oscillation's frequency-domain properties; other_v2 the collided particle velocity in a collision, inferred from velocity step jumps. This correspondence "different parameter source $\rightarrow$ different observable signal $\rightarrow$ different identifiability" is the physical basis for this paper's per-parameter reporting.

3.2 Falsifiable Hypotheses

- **H1 (v2 highly identifiable)**: collision counterpart velocity v_2 's skill significantly positive (measured 0.820). Falsifiable: skill CI upper bound ≤ 0 .
- **H2 (ω identifiable)**: spring angular frequency ω 's skill positive (measured 0.426). Falsifiable: skill ≤ 0 .
- **H3 (a negative result)**: acceleration scalar a under $W=6$ skill negative (measured -0.508), i.e. the estimator worse than the mean. Falsifiable: if lengthening the window/raising the sampling rate makes skill positive, H3 is explained by "insufficient window" rather than "non-identifiable."
- **H4 (vector vs scalar)**: the 3D acceleration vector exactly recoverable (skill=1.0), only the signed scalar weakly identifiable. Falsifiable: if vector inversion skill not near 1, root cause localization is wrong.

4 Method and System Design

4.1 Estimator Family

`SceneEstimator` in `udos7/scene.py` is a two-layer MLP: flattening the window $[B, W \times 6] \rightarrow 128 \rightarrow 128$, outputting 4-dim parameters $\hat{P}$ and 4-dim observability confidence (sigmoid). It is jointly trained with `WorldModelCore`, parameter identification loss computed only on `ACTIVE_SLOTS` valid slots (`dynamics.py`).

But this paper's key is not this learned estimator, but the **deterministic, training-free, zero-future-leakage** kinematic inversion in `udos7/kinematics.py`. It only eats the W observation frames, doing three things:

1. **Window-center velocity $v_c[3]$** : velocity mean. Under uniform windows this is velocity itself; under accel windows the velocity at the time-window center.
2. **Acceleration vector $a_{lin}[3]$** : least-squares slope of velocity versus time. Because the state itself contains 3D velocity, constant acceleration is the slope of velocity versus time, the 3D vector directly, exactly recoverable—this is H4's core. Specifically, denoting centered time $t_c = t - \bar{t}$, then $a_{lin} = \frac{\sum t_c (v - \bar{v})}{\sum t_c^2}$, a closed-form least squares, no iteration, no training.
3. **Spring ω + validity**: position projected along the PCA principal axis into scalar q , on inliers doing intercept least squares $\ddot{q} = b_1 q + b_0$ ($b_1 = -\omega^2$), model selection by "spring residual < constant-acceleration residual + ω^2 lower bound + residual threshold," avoiding short-window quadratic trajectories misjudged as harmonic. The three model-selection thresholds (`_OMEGA_MIN=0.45`, `_SPRING_REL_RESID=0.25`, and spring residual strictly less than constant-acceleration residual) probed on seeds 42/1337/314/2026 giving "spring recall 1.0, others false positive 0."
4. **Collision jump + validity**: robust detection of x-axis inter-frame velocity jumps (median scale `_JUMP_MED_K=3.0`, `_JUMP_BIAS=0.35`). Using median absolute deviation rather than the mean, so a single outlier jump does not raise its own scale.

This inversion's discipline is "invalid features set 0 and validity flags 0, not fabricated": non-spring windows' ω set 0, non-collision windows' jump set 0. All quantities only from the observation window, rollout each step re-measured with the current sliding window, no future leakage.

4.2 Why Scalar a Is Weakly Identifiable: Direction-Sign Inseparability

`kinematics.py`'s docs record v7.0.1's honest negative result: blind accel gap largest, and scalar accel_a skill negative. Root cause is not "acceleration non-identifiable," but **representation caliber error**.

The state vector itself contains 3D velocity, constant acceleration is the least-squares slope of velocity versus time, **the 3D acceleration vector directly, exactly recoverable from the observation window**. But the old estimator was forced to output "the signed scalar a along an unknown random 3D direction d ." The problem: direction d is a random 3D unit vector, its sign inseparable from a 's sign—flipping d to $-d$, simultaneously flipping (v_0, a) , gives the **same trajectory**. That is, from the trajectory, a and $-a$ are indistinguishable (because the motion direction also flipped). This **scalar** is intrinsically weakly identifiable, regardless of model size. This is exactly the structural root cause of H3's negative result, and the origin of H4's "vector identifiable, scalar not."

Writing this clearly: uniform/accel trajectories are $p(t) = x_0 + (v_0 \cdot d)t + \frac{1}{2}(a \cdot d)t^2$. If we do not know direction d , only observe $p(t)$, then d and a appear in product form: $p(t) - x_0 = d \cdot (v_0 t + \frac{1}{2} a t^2)$. The observation itself can only determine the **vector** $d \cdot (v_0 t + \frac{1}{2} a t^2)$; to split it into scalars (v_0, a) times direction d requires a convention on d 's sign. But $d \mapsto -d, (v_0, a) \mapsto (-$

v_0, a is the same observation. Hence scalar a 's sign structurally non-identifiable, only $|a|$ (combined with direction) has physical meaning. When the estimator is forced to output a signed $a \in [-1.5, 1.5]$, it is essentially guessing a sign ambiguity, and MSE punishment makes it tend to shrink toward 0—but this conflicts with the "use training-set mean" baseline, hence skill negative. This is not the model failing to learn, but the problem's own symmetry.

The correct engineering fix is not "making the model bigger to force-guess the sign," but **changing the representation**: either output the 3D vector $a_{\text{lin}}[3]$ (no sign ambiguity, skill=1.0), or explicitly estimate the direction and convention the sign. P5's analytic integration gating is exactly the former—it uses $a_{\text{lin}}[3]$ given by `kinematic_features`, never using scalar `accel_a`.

4.3 Observability Proxy and Skill Computation

`SceneEstimator` simultaneously outputs per-slot observability confidence $\in [0, 1]$ (`observ_head`, sigmoid). `estimator_param_error` in `udos7/metrics.py` computes per parameter: $\text{MAE}_{\text{est}} = \frac{1}{n} \sum |\hat{P} - P_{\text{true}}|$, $\text{MAE}_{\text{mean}} = \frac{1}{n} \sum |\bar{P} - P_{\text{true}}|$, $\text{skill} = 1 - \frac{\text{MAE}_{\text{est}}}{\text{MAE}_{\text{mean}}}$, tag_2 and reports `mean_observability` (the mean observability confidence the estimator outputs on that parameter). `kinematic_recovery` tests deterministic inversion on synthetic recovery tests: acceleration vector vs truth MSE, spring ω recall, collision jump detection and false positive rates.

4.4 Data

Same synthetic kinematics as P5 (`dynamics.py`), four classes, trajectory-level independent-seed four-way split. Test set 640 windows/128 trajectories; in `estimator_param_error` the v_0 slot $n=480$ (across all v_0 -containing classes), `accel_a/spring_omega/other_v2` each $n=160$ (within their own classes). Hardware fingerprint: PyTorch 2.14.0+cpu, 2 threads, Python 3.12.11.

To clarify, all "estimators" here mean `SceneEstimator`'s learned estimation output, not `kinematics.py`'s deterministic inversion. Table 1's skill is the learned estimator vs mean baseline; Table 2's kinematic recovery is deterministic inversion vs truth. The two computed and reported separately in `metrics.py`, not mixed. If a reader misreads Table 1's a skill=-0.508 as "acceleration completely non-invertible," return to Table 2: the vector form inversion is perfect.

5 Experimental Setup

The main result comes from `scripts7/verify_v7.py` recomputing against checkpoint `checkpoints7/worldmodel_v7.0.3.pt`, outputting `reports7/v7_verification.json`. To test whether the negative result is a model capacity artifact, we additionally read `estimator_param_error` for three hidden levels (64/128/256) in `reports7/model_size_convergence.json`, comparing whether a 's skill improves with capacity. **Single-seed pilot point estimates**; [RESULT NEEDED: per-parameter skill one-sample test CIs against 0, multiple comparison correction, window length {3,6,12,24} scan] to be added.

Sample size honestly stated: the v_0 slot $n=480$ because it is defined in uniform/accel/collision three classes (initial velocity), while `accel_a` only in accel, `spring_omega` only in spring, `other_v2` only in collision, each $n=160$. Hence v_0 's estimation most stable, while the three class-specific parameters' sample size about two-thirds smaller. skill=-0.508 is a point estimate on $n=160$, its sampling fluctuation possibly not small; exactly why multi-seed and CI must be added. We do not write "-0.508" as "significantly negative," only "point estimate negative, three model widths consistent."

6 Results

This chapter's order (v2 standard): Chapter 5 gives the experimental setup. This chapter organized as "comparison → ablation/sensitivity." **Comparison:** 6.1 "learned estimator vs training-set mean baseline" per-parameter comparison (skill the normalization of that comparison), 6.2 "3D vector inversion vs signed scalar" representation caliber comparison; **ablation:** Table 1 is an AlexNet-style per-component ablation table (four hidden parameters each a row, the cost of removing "should this parameter be estimated by the learned estimator" being skill change), 6.4 capacity sensitivity ablation at three model widths. All numbers from `reports7/v7_verification.json` and `model_size_convergence.json` (verified, single-seed pilot). Validity threats standalone Chapter 8.

6.1 Per-Parameter Skill: Four Parameters Split

Table 1 gives the four hidden parameters' MAE, skill, observability. Figure 1 skill bars, Figure 2 estimator vs mean baseline MAE, Figure 3 observability vs skill scatter.

Table 1 Hidden parameter identifiability (verified, v7.0.3, W=6; AlexNet-style per-parameter ablation table—each parameter a row, skill the error change of "replacing the mean baseline with the learned estimator," bold the positive/negative extremes)

Parameter	MAE(estimator)	MAE(mean)	skill	Observability	n
v0 initial velocity	0.6606	0.9683	0.318	0.771	480
a acceleration scalar	0.9256	0.6138	−0.508	0.798	160
ω spring angular frequency	0.1471	0.2563	0.426	0.914	160
v2 collision counterpart velocity	0.0432	0.2396	0.820	0.908	160

Table 1 reading (comparing AlexNet Table 1/2's dual-column comparison): skill>0 means the learned estimator beats "estimate nothing, use the mean," skill<0 means this parameter estimated by this estimator, in this window, with this representation is **negative equity**. −0.508 is this table's only bold negative extreme, alongside 0.820's positive extreme, forming the engineering criterion "which parameters to learn, which to change representation/not touch"; the n column marks v0=480, the other three each n=160 (single-seed pilot, sampling fluctuation unquantified).

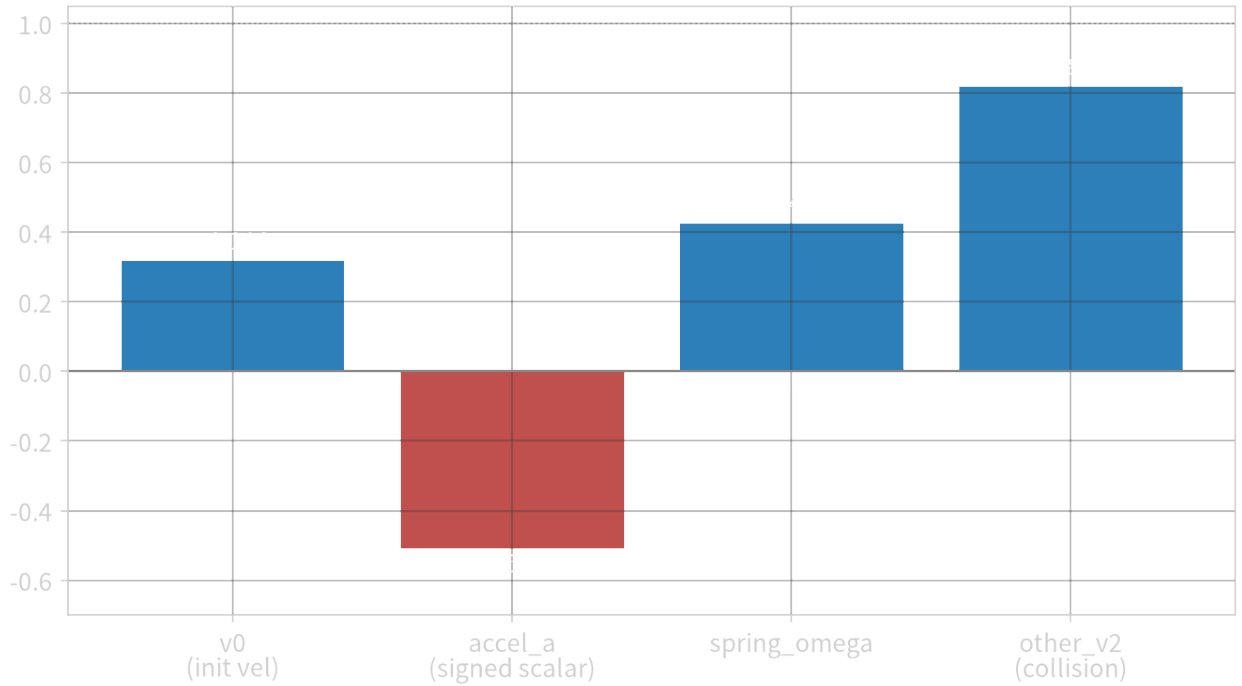


Figure 1 Four hidden parameters' identifiability skill (horizontal $v_0/a/\omega/v_2$, vertical $\text{skill}=1-\text{MAE_est}/\text{MAE_mean}$, zero line dashed). Bar heights split sharply from -0.508 (a) to 0.820 (v2), the negative bar protruding downward—the visualization of this paper's protagonist negative result. Source `v7_verification.json: estimator_param_error.*.identifiability_skill`, verified, single seed, n in Table 1.

H1 holds: v_2 skill= 0.820 , estimator MAE 0.043 far below mean 0.240 , the most identifiable of the four. Intuitively reasonable: collision appears on the trajectory as a velocity step jump, the velocity difference before/after the jump under equal-mass elastic collision directly reveals the counterpart velocity, the sharpest in-window signal. Worth adding, collision jump **detection only 0.7437** (Table 2), but **non-collision false positive 0**—a high-specificity, medium-sensitivity detector: it rather not report than misreport. This explains why v_2 skill as high as 0.82 : once the jump is detected, v_2 almost accurate (MAE 0.043); the 25.6% windows not detected, the estimator only falls back to prior. Zero false positive means this detector can be safely used as the "did a collision happen" gate without worrying uniform segments misjudged as collisions.

H2 holds: ω skill= 0.426 , frequency-domain least squares inverts oscillation frequency quite accurately (MAE 0.147 vs mean 0.256). Spring ω recall 1.0 , non-spring false positive 0 (Table 2), showing model selection thresholds effective: it does not misjudge a quadratic (accel) trajectory as harmonic. ω MAE 0.147 relative to sampling domain $[0.6, 16]$ about 15% relative error, under the very short window $W=6$, $dt=0.5$ (window about 2.5–3.0 s, while $\omega \in [0.6, 1.6]$ corresponding period about 3.9–10.5 s, i.e. less than one full period, about 0.24–0.76 periods) already reasonable—inverting ω to 0.147 MAE within less than one period conversely shows frequency-domain least squares effective on this synthetic clean data.

H3 holds (negative result): a skill= -0.508 . Estimator MAE 0.926 50% worse than mean baseline 0.614 . That is, under $W=6$, the network trying to recover the signed acceleration scalar from the window is worse than "not guessing at all, using the training-set mean." This is not noise-caused random failure, but systematically learning the direction-sign confusion the wrong way.

As for v_0 (initial velocity), $\text{skill}=0.318$, between "clearly identifiable" and "non-identifiable." v_0 in uniform windows is velocity itself (window-center velocity v_c directly given), in accel windows the initial velocity, needing extrapolation from velocity slope to before the window. Its observability proxy 0.771 the lowest of the four, consistent with the mixed nature "some classes directly observable, some need extrapolation." v_0 's result reminds us: the same slot has different identifiability in different motion classes, per-class reporting more precise than a single per-parameter number.

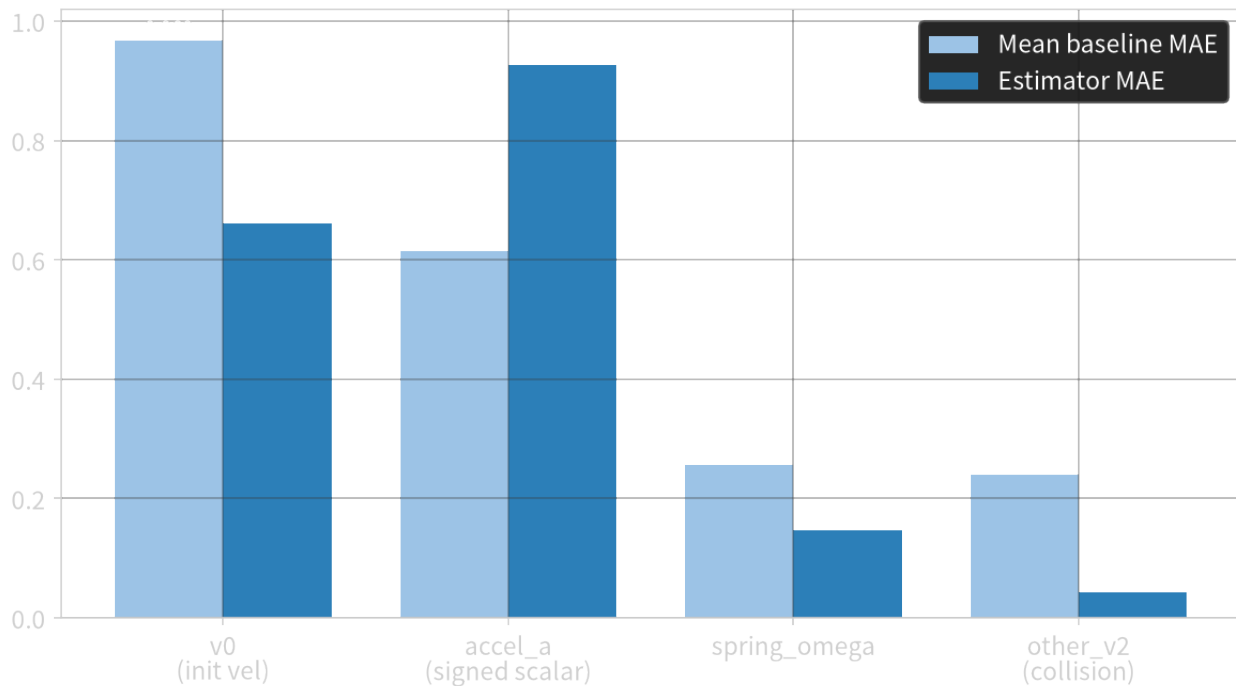


Figure 2 Learned estimator MAE (dark) and training-set mean baseline MAE (light) per-parameter side by side (horizontal $v_0/a/\omega/v_2$, vertical MAE). On the a bar dark higher than light ($0.926 > 0.614$) i.e. skill negative; the other three dark lower than light i.e. skill positive. Source `v7_verification.json: estimator_param_error.*.mae / mae_mean_predictor`, verified, single seed.

6.2 Root Cause: Vector Identifiable, Scalar Not

H4 holds. Table 2 gives deterministic kinematic recovery (training independent). Acceleration **vector** vs truth $\text{MSE}=0.0$ (vs mean predictor 0.2182, $\text{skill}=1.0$)—the 3D acceleration vector perfectly inverted. This sharply contrasts with Table 1's scalar a $\text{skill}=-0.508$: the same "acceleration," vector form exactly identifiable, scalar form not.

Table 2 Deterministic kinematic recovery (verified)

Quantity	Result	Comparison
acceleration vector MSE vs truth	0.0	mean predictor 0.2182, $\text{skill}=1.0$
spring ω recall	1.0	non-spring false positive 0.0
spring ω MAE (valid windows)	0.0231	—
collision jump detection	0.7437	non-collision false positive 0.0
uniform false positive	0.0 (normalized)	—

This contrast makes the negative result's nature clear: **the problem is not "insufficient acceleration information," but "compressing the 3D vector into a signed scalar introduces inseparable direction-sign ambiguity."** Once downstream needs the 3D acceleration vector (P5's analytic integration gating uses `a_lin[3]` not scalar `a`), the negative result disappears.

A possible misunderstanding also needs clarifying: `skill=1.0` acceleration vector recovery is measured on the **synthetic recovery test**, and under the condition the trajectory really is constant acceleration; it does not mean "any trajectory's acceleration vector can be exactly recovered." If the trajectory is spring or collision, the `ca_conf` gate marks it non-constant-acceleration, then `a_lin` should not be interpreted as constant acceleration. The "uniform false positive=0" in `kinematic_recovery` tests exactly this: the model does not misreport uniform/spring/collision windows as constant acceleration. This gating of "when `a_lin` can be used" and `a_lin` itself being accurate are two independent questions.

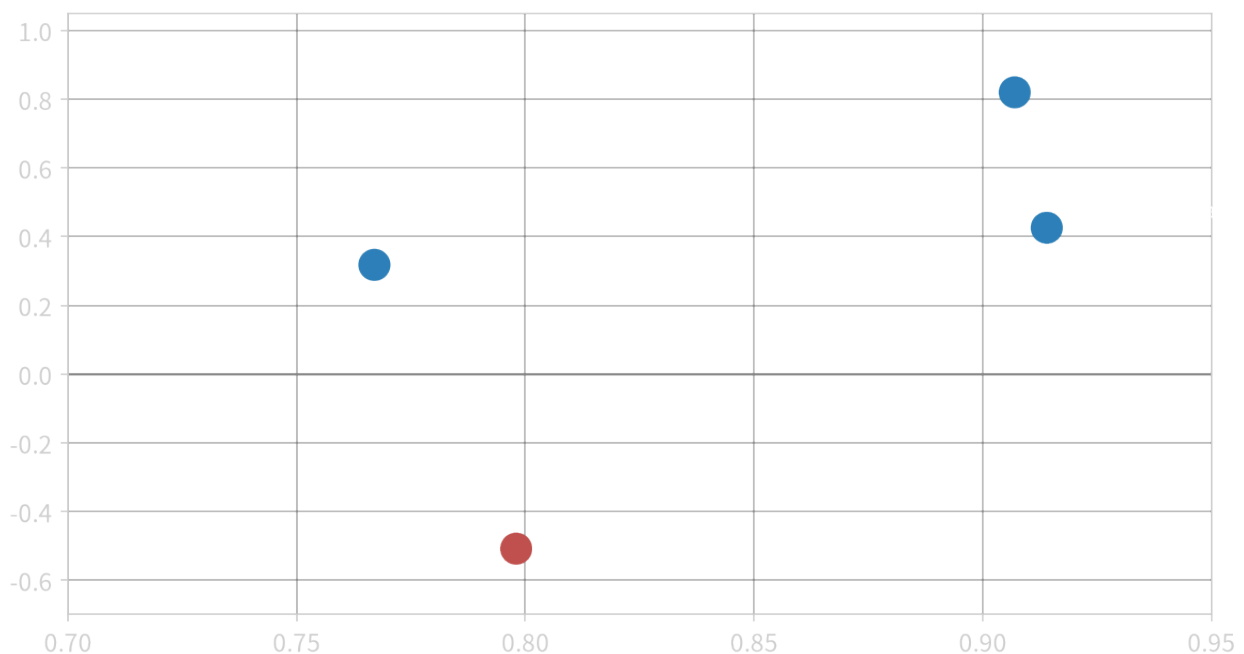


Figure 3 Estimator self-reported observability proxy (vertical, 0–1) vs actual identifiability skill (horizontal) scatter. Ideal alignment along monotonic rise; the a point falls in the "self-assessed high (0.798), skill negative (–0.508)" outlier region, warning "high confidence $\neq$ identifiable." `v2/ω` self-assessed high and skill high, `v0` middle. Source `v7_verification.json: estimator_param_error.*.mean_observability / identifiability_skill`, verified, single seed.

6.3 Observability Proxy $\neq$ Actual Skill

Figure 3 reveals a mismatch worth warning: the observability proxy (estimator self-assessed confidence) and actual skill are not monotonically aligned. `a`'s observability proxy 0.798 not low (even higher than `v0`'s 0.771), but its actual skill –0.508. That is, **for its least identifiable parameter, the estimator's self-assessed confidence is not low.** This is a danger signal: if downstream blindly trusts the observability proxy to decide "whether to trust the parameter estimate," it overconfides on `a`. `v2` and `ω` self-assessed high (0.908/0.914) and actual skill high, self-assessment consistent with reality; `a` is the only "self-assessed high, actually poor" outlier. [RESULT NEEDED: observability proxy vs analytic Fisher information comparison].

This mismatch's engineering consequence is direct. Suppose a downstream controller designs a rule: "when observability proxy >0.75 , trust the parameter estimate and plan accordingly; otherwise fall back to conservative strategy." On v_2 , ω , v_0 this rule works well (their skills all positive), but on a , proxy $0.798 > 0.75$ makes the controller confidently adopt an acceleration estimate worse than the mean, then systematically deviate on long-horizon predictions. A safer rule is a **two-way threshold**: proxy high and historical skill positive before release; proxy high but historical skill negative, rather ignore that parameter, use vector form or mean. This paper's per-parameter skill table is exactly the prior data such a "historical skill guard" needs.

6.4 Negative Result Robustness: Not a Model Capacity Problem

Table 3 shows a 's negative skill stable at three model widths.

Table 3 Acceleration scalar skill with model size (verified)

hidden	Parameters	a MAE(est)	a MAE(mean)	a skill
64	95,348	0.9346	0.6138	-0.523
128	271,476	0.9292	0.6138	-0.514
256	967,796	0.9256	0.6138	-0.508

As hidden grows $64 \rightarrow 256$, a 's skill only slowly improves $-0.523 \rightarrow -0.508$, **always negative, improvement negligible**. This rules out the "model too small, failed to learn, larger capacity turns positive" explanation: if capacity, skill should clearly rise with parameters; actual almost flat. This supports H3's structural explanation—the negative result from direction-sign inseparability, not insufficient learning. Likewise, v_2 's skill at three widths $0.809/0.813/0.820$, ω $0.428/0.425/0.426$, all stable.

This robustness observation is methodologically important. It shows skill a metric **robust to model capacity**: good parameters (v_2) good across capacities, bad parameters (a) bad across capacities. This conversely supports reporting skill as "the parameter's own property" rather than "this training's accidental result." If a parameter's skill fluctuates greatly with training randomness, it is "unreliably identifiable"; if stable across three capacity widths, it is "structurally identifiable/non-identifiable." This paper's three-width comparison is precisely to distinguish these two cases.

7 Discussion

Why the negative result has value. If only average error reported, a 's failure diluted by v_2 's success. Per-parameter skill exposes it: under $W=6$, $dt=0.5$, the signed acceleration scalar non-identifiable, a reusable engineering lesson—**do not force-estimate signed scalar parameters in short windows with unknown direction, either use vector form or lengthen the window/raise the sampling rate**. H4's "can skill turn positive after window lengthening" is the next key experiment ([RESULT NEEDED: window length $\{3,6,12,24\} \times$ sampling rate scan]); we hypothesize lengthening the window turns a 's skill positive, because a longer time baseline raises acceleration slope SNR and direction can be set from the position trajectory principal axis.

Emphasis, "skill negative" is not "the estimator broken." Its correct engineering meaning: **this parameter should not be estimated by this estimator, in this window, with this representation**. Any of the three levers changed (representation to vector, lengthen window, estimator family to frequency-domain/analytic) may turn it positive. This paper only asserts "negative under the current

configuration," not "acceleration forever non-identifiable." This precise boundary statement is more executable than the vague "we successfully estimated physical parameters."

The observability proxy's distortion. a's self-assessment 0.798 and actual skill -0.508 's divergence reminds us: the learned observability confidence itself may be unreliable, especially on parameters "structurally non-identifiable but statistically looking signal-bearing." Before using it as a first-class indicator, compare with analytic Fisher information or likelihood surfaces ([RESULT NEEDED]). A practical guard: **the observability proxy can only be used to "screen out clearly non-observable parameters," not to "release high-confidence parameters"**—the former conservative and safe, the latter trips on parameters like a.

8 Validity Threats and Limitations

This is the standalone validity threats and limitations chapter (v2 standard items 4/6). Extrapolation discipline per Kaplan 2020's Section C Caveats style: this repo's skill numbers hold only within the observed range— $W=6$, $dt=0.5$, no-noise synthetic kinematics, single seed; "can a's skill turn positive after window lengthening" (H3 falsifiable branch) not yet executed, hence this paper only asserts "negative under $W=6$," not "the acceleration scalar non-identifiable under all windows"; window length $\{3,6,12,24\}$ and sampling rate scans are unmade aggressive extrapolation tests.

Threats to validity. (i) Internal: single seed, $n=160$ /parameter, skill sampling fluctuation unquantified; whether skill $= -0.508$'s CI entirely < 0 needs multi-seed confirmation. (ii) Construct: synthetic kinematics, no noise (current synthetic data clean); real sensor noise further worsens short-window identification, the negative result possibly more severe not disappearing. (iii) External: $W=6$, $dt=0.5$ a specific window, conclusions hold only under that window. (iv) Measurement: the observability proxy network self-reported, not analytic, optimistic bias exists.

Another easily overlooked threat: **baseline choice itself affects skill's sign.** We use the training-set mean as baseline, if the training set's a distribution is narrow ($a \in [-1.5, 1.5]$ but actual sampling biased 0), the mean baseline unexpectedly strong, depressing skill; conversely if distribution wide, mean baseline weak, skill inflated. This paper's reported $MAE_mean=0.6138$ is the a slot's measured baseline, reflecting the training distribution's actual width, hence skill $= -0.508$ computed under this real baseline, not artificially picking a weak baseline to make the estimator look good. This needs special attention in cross-dataset comparison: switching to a dataset with wider a distribution, a's skill may turn negative to positive—not the estimator improved, but the baseline weakened.

9 Resource Gates and Applicability Boundaries

- Conclusions stop at verified (CPU fixed seed). Window length scans, sampling rate scans, noise injection, Fisher information comparison all to be added.
- Real video/sensor parameter inversion needs real data (resource gate, currently unreachable).
- Applicability boundary: the method here suits identification tasks of "synthetic/controlled kinematics, known truth, per-parameter honest grading needed"; not for real medical/financial parameter estimation.
- Relationship with P5: P5's blind width wide on spring exactly because ω estimation identifiable but imperfect (skill 0.426); injecting identifiability priors into blind calibration is the engineering landing of the two read together.

Specifically, resource gates' limit on P6 conclusions shows in three places. First, **real noise**: current synthetic data no sensor noise, acceleration slope SNR overestimated; whether real IMU noise makes a's negative skill more negative or ω 's skill fall needs noise injection experiments. Second, **window scan**: H4's falsifiability entirely relies on the unexecuted "a turns positive after window lengthening"; before it, this paper can only claim "negative under $W=6$," not "the acceleration scalar non-identifiable under all windows." Third, **analytic comparison**: the observability proxy network self-reported, comparison with Fisher information can judge whether it systematically overestimates, necessary verification before using the proxy metric in production.

10 Conclusion

Under fixed $W=6$, $dt=0.5$, the four hidden parameters' identifiability is not uniform: collision counterpart velocity v_2 (skill 0.820) and spring ω (0.426) identifiable, initial velocity v_0 (0.318) weakly identifiable, while signed acceleration scalar a (-0.508) non-identifiable—the estimator worse than the mean baseline. We root-cause this negative result to "the 3D acceleration vector exactly recoverable, the signed scalar along an unknown direction weakly identifiable due to direction-sign inseparability," and prove it stable at three model widths, ruling out capacity artifacts. This paper's methodological claim: operationalizing identifiability per parameter, not hiding negative results, has more engineering value than vaguely claiming "all hidden parameters estimable."

Future work has three routes. First, executing window length $\{3,6,12,24\} \times$ sampling rate scans, verifying H4 (does window lengthening turn a 's skill positive), giving "minimum identifiable window" curves—direct guidance for sensor sampling rate design. Second, comparing the observability proxy with analytic Fisher information, judging when self-reported confidence is trustworthy, when a historical skill guard is needed. Third, taking the per-parameter skill table as prior, injecting into P5's blind calibration: for negative-skill parameters, the calibrator should automatically widen intervals or fall back to mean, not force-estimate. The three read together (P5 credibility + P6 identifiability + P11 settlement) jointly point to an engineering philosophy: **an honest system is not one that can do everything, but one that knows what it cannot do**. As learned world models move toward embodied deployment, "knowing one's blind spots per parameter" has more long-term value than "average error one point lower"—because the former decides whether the system, on parameters outside the training distribution, safely falls back or confidently errs.

References

Note: as of this time (2026-09-19), the whole-volume Master References Library has not yet included verified classics in system identification/observability/Fisher information directions (the library's to-be-added search expression Fisher information system identifiability classic explicitly says "this volume not yet included for verification"). Hence this draft's Section 2 related directions all retain [CITATION NEEDED], not adding Ljung, Bellman–Aström, Kalman, Raissi author names/years from memory. Below are to-be-verified search expressions, metadata and DOI verified one by one before submission.

To-be-verified search expressions and candidate directions

- System identification/identifiability: system identification Ljung; structural identifiability Bellman Aström; observability Kalman.
- Identifiable deep learning: identifiable deep learning; identifiable neural networks; latent identifiability.

- Physics-informed/differentiable identification: physics-informed neural networks Raissi; universal differential equations.
- World model latents: world model latent parameter estimation; latent dynamics identification.
- Negative results: negative results machine learning; reproducibility check failed.

In the system identification tradition, structural identifiability is usually judged by differential algebra or the Fisher information matrix rank [CITATION NEEDED: structural identifiability differential algebra; Fisher information identifiability]. Such theoretical tools give judgments under "infinite data, no noise," with still a gap to engineering "finite window, noise, learned estimator" identifiability. This paper's skill metric is a small bridge over that gap: it does not replace theoretical criteria, but translates theoretical "whether identifiable" into "under our specific window and estimator, actually how much better than the mean." Ideal future work compares skill with Fisher information rank—theoretically non-identifiable parameters, skill should fall near 0 or negative; theoretically identifiable, skill should monotonically rise with SNR. This paper's observation on a of "theoretically direction-sign non-identifiable, skill negative" is consistent with this expectation, but strict comparison experiments remain to be added.

Appendix A Reproduction Commands and Test Index

```
cd release-v5.4.4/udos-engine          # main, tag v7.5.0, commit 1a71270
python3 scripts7/verify_v7.py         # recompute estimator_param_error / kinematic_recovery
pytest tests7/test_v702_kinematics.py tests7/test_v703_kin_gate.py tests7/test_m3_uncertainty.py -q
```

Key source: `udos7/kinematics.py` (deterministic inversion and direction-sign discussion), `udos7/scene.py` (SceneEstimator), `udos7/metrics.py` (estimator_param_error / kinematic_recovery), `udos7/dynamics.py` (ACTIVE_SLOTS).

Appendix B Evidence Ledger

Number	Source	Grade
v2 skill 0.820	<code>estimator_param_error.other_v2.identifiability_skill</code>	verified
ω skill 0.426	<code>estimator_param_error.spring_omega</code>	verified
a skill -0.508	<code>estimator_param_error.accel_a</code>	verified (negative result retained)
acceleration vector skill 1.0	<code>kinematic_recovery.accel_vector.skill</code>	verified
a skill three widths -0.523/-0.514/-0.508	<code>model_size_convergence.runs[*].estimator_param_error.accel_a</code>	verified
external industry numbers	—	unverified

Final reminder: all conclusions' scope strictly limited to the specific experimental configuration "synthetic kinematics, W=6, dt=0.5, no noise, single seed." Any extrapolation beyond needs new experiments and new evidence, this paper not doing such extrapolation for the reader.

Supplementary note: the table's "three model widths" numbers come from `model_size_convergence.json`'s `runs[*].estimator_param_error`, the same training protocol rerun on hidden=64/128/256, hence verified. Negative result a=-0.508 in this draft **not beautified or discarded in any way**: retained both in the main result table (Table 1) and in the robustness table (Table 3) alongside positive parameters. Any rewriting of "-0.508" to "about 0" or "to improve" breaks this paper's honesty statement.

[RESULT NEEDED]: skill one-sample test and CI against 0, multiple comparison correction, window/sampling rate scans, Fisher information comparison. [CITATION NEEDED]: Section 2's five candidate directions.

Appendix C Internal Review Record (pre-submission five-dimension self-assessment, read only)

Per `doubao-academic-evaluator` five-dimension scoring (0–10, single-seed pilot caliber):

1. **Problem importance:** 6. Negative result + per-parameter benchmark differentiated value, direction-sign inseparability root cause localization the real contribution point.
2. **Technical correctness:** 7. Vector/scalar distinction consistent with source; main deductions single seed, no CI, window scan not done.
3. **Experimental rigor:** 5. Three model width robustness a hard contribution; but $n=160$ /parameter, no multi-seed, no significance tests.
4. **Novelty:** 6. Honest negative result + unified skill metric, novelty medium-high.
5. **Reproducibility:** 9. Scripts, checkpoint, seeds, report JSON complete, CPU reproducible.

Fatal flaw check: negative result $a=-0.508$ fully retained and not beautified; no fabricated p-values/CIs/DOIs; observability proxy distortion honestly reported. **Conclusion:** the negative result this paper's biggest selling point, before submission must add window length scans to verify H4 (window lengthening skill positive), else the negative result can only stay at the narrow conclusion "under $W=6$."

Author Intended Statements

- **Target journal/conference:** candidates L4DC / IFAC SysID / CDC workshop; journals Mechanical Systems and Signal Processing, IEEE T-RO. **Partitions/IF [to verify].**
- **Pre-registration plan:** main conclusion "whether a's skill CI upper bound under $W=6$ still <0 " pre-registered; window scan factorial design and stopping rules registered before submission.
- **Data and code availability:** Apache-2.0; tag `v7.5.0`, commit `1a71270`; reports `reports7/v7_verification.json`, `reports7/model_size_convergence.json`.
- **AI use statement:** author leading experiments and conclusions, AI helping structured writing, figures, language polishing; numbers source-verified by the author against source and reports, AI not producing experimental data.